PCD Assignment 02

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Image Enhancement on a Blurred Image Using Unsharp Masking

1. Introduction

Image enhancement is a fundamental process in digital image processing that aims to improve the visual quality of an image or make important image information easier to observe. Images can suffer from degradation such as blur, low contrast, excessive brightness, insufficient brightness, and noise. In this experiment, a blurred image is selected as the enhancement case. Blur reduces the visibility of edges and fine details, so a sharpening technique is used to improve the appearance of the degraded image.

The method used in this experiment is Unsharp Masking. This method emphasizes high-frequency information such as object boundaries, lines, and texture details. A basketball court photograph is used as the original reference image because it contains many useful structures for observing blur and sharpening, including court lines, tree branches, fences, walls, and other edges.

2. Methodology

The experiment starts by loading the original RGB image. Gaussian Blur with sigma = 2.0 is then applied to create a controlled blurred image. This step simulates a loss of sharpness while preserving the original image as a ground-truth reference. Keeping the original image allows the degraded and enhanced results to be evaluated objectively.

For enhancement, the blurred image is smoothed again using Gaussian Blur. The high-frequency detail is represented by the difference between the blurred image and this smoother version. The detail is then added back using a sharpening amount of 2.5. This operation is an implementation of Unsharp Masking and increases local edge contrast, making boundaries appear clearer.

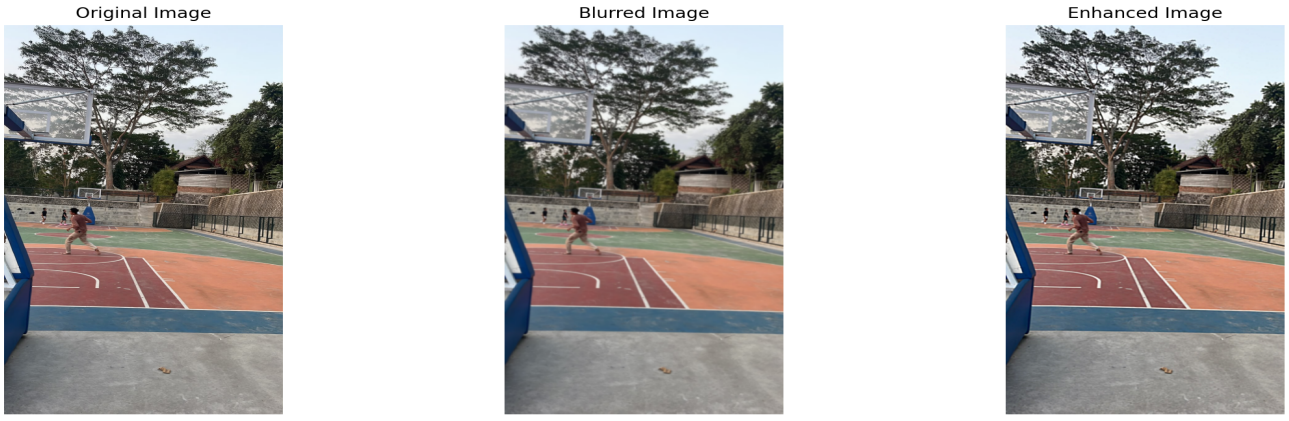


Figure 1. Visual comparison of the original, blurred, and enhanced images.

3. Result and Analysis

The Gaussian-blurred image shows a noticeable reduction in edge sharpness. Fine structures in the tree branches, fence, court markings, and background objects become softer. After Unsharp Masking is applied, these structures become more visible because the local contrast around edges is increased. The enhanced result is not identical to the original image, because sharpening cannot perfectly recover

information that has already been removed by blurring, but it improves the perceived sharpness.

Image	MSE	PSNR
Blurred	682.2215	19.7915 dB
Enhanced	502.6888	21.1178 dB

The blurred image produces an MSE of 682.2215 and a PSNR of 19.7915 dB. After enhancement, the MSE decreases to 502.6888, while the PSNR increases to 21.1178 dB. A lower MSE indicates a smaller average squared difference from the original image, while a higher PSNR generally indicates better similarity to the reference image. Therefore, the numerical results support the visual observation that the sharpening operation improves the degraded image for this experiment.

It is important to note that quantitative metrics do not always represent perceived visual quality perfectly. Sharpening can make edges look clearer while also changing pixel values. For this reason, both visual comparison and objective metrics are considered in the analysis.

4. Conclusion

This experiment successfully implements image enhancement for a blurred image using Unsharp Masking. Gaussian Blur is first used to create a degraded image, and Unsharp Masking is then applied to improve edge visibility and perceived sharpness. The result shows clearer boundaries than the blurred input. The MSE and PSNR measurements also improve after enhancement, indicating that the enhanced image is closer to the original reference than the blurred image. Overall, Unsharp Masking provides a simple and effective approach for enhancing blurred images, although it cannot completely reconstruct every detail lost during the blurring process.